INCH-POUND
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30 September 1991
SUPERSEDING
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10 May 1974

MILITARY SPECIFICATION

TETRAETHYLENEPENTAMINE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of tetraethylenepentamine.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810
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STANDARDS

FEDERAL

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 86	-	Standard Method for Distillation of Petroleum Products
ASTM E 133	-	Standard Specification for Distillation Equipment

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.7.

3.2 Chemical and physical analysis. The chemical and physical analysis of the material shall be as specified in table I.

TABLE I. Chemical and physical analysis.

Characteristic	Minimum	Maximum
Distillation temperature, °Celsius(C) at 760 millimeters (mm) barometric pressure:		
- for 5 milliliters (ml) of distillate	310	-
- for 50 ml of distillate	320	-
Chlorides, percent	-	0.5
Moisture, percent	-	1.0

3.3 Material safety data sheets. A material safety data sheet shall be prepared in accordance with the requirements of FED-STD-313 (see 6.6).

3.4 Workmanship. The tetraethylenepentamine shall be uniform in quality, free from foreign materials, and manufactured under controlled conditions and procedures in the industry.

WARNING

This is a hazardous chemical (liquid). The vapors may be toxic and can cause skin irritations. Decomposition produces nitrogen oxides and other chemicals which may cause health problems depending on the duration and extent of exposure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.6.2).

4.3 First article sample. The first article sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of tetraethylenepentamine. The first article sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified, the Government will witness the examinations and tests for first article sample acceptance at the contractor's plant. First article samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the first article sample has been obtained (see 3.1).

4.4 Lot. A lot shall consist of material produced at one plant with no change in formulation or process. If manufacture is by batch process, each batch shall constitute a lot (see 6.4).

4.5 Acceptance sampling. The number of containers to be chosen at random for acceptance sampling shall be equal to the square root of the total number of containers in the lot. If the number thus obtained is not a whole number, the number of containers to be sampled shall be increased to the next higher whole number. In no case, however, shall the number of containers to be sampled be less than seven (unless there are less than seven containers in the lot, in which case each container shall be sampled) (see 4.2).

4.5.1 Primary sample. From each selected container, a sample shall be taken from three or more places throughout the container. The total weight of the samples taken from each container shall weigh at least 50 grams (g). Each sample thus taken shall be mixed thoroughly, placed in a clean, dry container, and labeled to identify the material name, original container designation, contract number, and lot number (see 4.2).

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4.5.2 Composite sample. Each primary sample shall be subdivided to prepare a composite sample (not in excess of 500 g). Primary material not used shall be returned to the primary sample container. After mixing the composite sample thoroughly, the composite sample shall be placed in a clean, dry container and sealed. The composite sample shall be identified with the material name, container designation, contract number, and lot number. All specified chemical and physical tests shall be made on this composite sample representing the lot. Failure of the composite sample to pass all of the tests herein shall reject the lot represented (see 4.2).

4.6 Classification of tests. Inspection and testing of the tetraethylenepentamine shall be classified as follows:

a. First article tests.

b. Quality conformance tests.

4.6.1 First article tests. First article tests shall be conducted only on the first article sample and shall consist of all the examinations and tests specified herein (see 3.1 and 4.2).

4.6.2 Quality conformance tests. Quality conformance tests for acceptance of the tetraethylenepentamine shall consist of the following examinations and tests listed in table II (see 4.2).

TABLE II. Quality conformance inspection.

Examination and test	Requirement paragraph	Test paragraph
Distillation temperature	3.2, table I	4.7.1
Chlorides	3.2, table I	4.7.2
Moisture	3.2, table I	4.7.3

4.7 Tests. The following procedures shall be used to determine that the requirements of this specification have been met. Any changes in the test procedures employed shall be subject to prior approval of the procuring activity. Unless otherwise specified, all tests shall be run in duplicate. The average of the two results shall be taken as the test result (see 3.2).

4.7.1 Distillation temperature. The distillation temperature shall be determined at 5 ml and 50 ml of distillate in accordance with ASTM D 86 (see table I).

4.7.1.1 Apparatus. Details and specifications for distillation apparatus are specified in ASTM D 86 and further described in ASTM E 133. The apparatus should be assembled as shown in figures 1 or 2 of ASTM E 133.

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4.7.1.2 Distillation procedure. The distillation procedure shall be in accordance with ASTM D 86 (see table I and 4.7.1).

4.7.1.3 Acceptance criteria. For the lot represented to pass the distillation temperature test, the values obtained for the distillation temperature at 5 ml and 50 ml of distillate shall be not less than specified in 3.2.

4.7.2 Chlorides. Introduce 10 ml of the sample into a 40-ml porcelain crucible by means of a transfer pipet. Evaporate the sample to dryness by heating slowly with a Fisher burner. (Take care to prevent loss of the sample by spattering.) Ignite the crucible to a dull red heat over a Fisher burner for 10 minutes, and allow to cool to room temperature. Transfer the crucible to one of two 600-ml beakers and add sufficient distilled water to cover it. To the other beaker add an equivalent amount of distilled water and reserve as the blank. Place the beakers on a 500-watt electric heater, heat the solutions to boiling, and allow the solutions to cool to room temperature. To each beaker add 6 to 8 drops of 1.0-percent alcoholic phenolphthalein indicator and 2 normal (N) nitric acid dropwise until the pink color just disappears. Add 1 ml in excess. Slowly add powdered commercially pure calcium carbonate until effervescence ceases. Pipet exactly 5 ml of standard 0.1N sodium chloride into each beaker. Add 5 ml of a 2.0-percent aqueous dextrin solution and 10 drops of 0.1-percent alcoholic dichlorfluorescein indicator. Stir the contents of the breaker vigorously with a glass rod and titrate with standard 0.1N silver nitrate to a salmon-pink endpoint. A blank using identical quantities of reagents is run simultaneously with the sample.

$$\text{Percent chlorides, by weight as sodium chloride} = \frac{(A - B)\ N \times 5.85}{10 \times C}$$

Where:

A = volume of 0.1N silver nitrate used to titrate the sample, ml

B = volume of 0.1N silver nitrate used to titrate the blank, ml

C = the specific gravity of the sample

10 = volume of the samples, ml

4.7.2.1 Acceptance criteria. For the lot represented to pass the chlorides test, the value obtained for the percent chlorides as sodium chloride shall be not greater than the value specified in 3.2.

4.7.3 Moisture.

4.7.3.1 Apparatus. The apparatus used for determination of moisture content of the sample shall be an Aquameter, Model KF-4, (Beckman Instruments, Inc., Fullerton, California), or an approved equivalent. The Aquameter shall

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be prepared for operation as described in the technical manual furnished by the manufacturer.

4.7.3.2 Reagents.

4.7.3.2.1 Karl Fischer reagent. Karl Fischer reagent must have a strength such that each ml of Karl Fischer reagent corresponds to 0.0014-0.0023 g of water. Dilute 750 ml of commercially available stabilized Karl Fischer reagent (with water equivalent of 0.005-0.007 g/ml) to 2000 ml with absolute methanol (0.1 percent water, maximum). Mix well and allow to stand overnight before use. Determine the water equivalent (A) of this solution as follows:

a. Use sodium tertrate dihydrate ($Na_2C_4O_6$ $2H_2O$), a primary standard (with a water content of 15.66%) for standardizing Karl Fischer reagent. If the water content value is in question, it may be determined by heating some of the salt at 150°C for 3 hours. Should the value (as determined) differ from the theoretical value of 15.66, then the experimental value shall be used in the determination of water equivalent (A) of the Karl Fischer reagent; instead of the 15.66 in the formula below, the factor should be 10P where P is percentage moisture (as determined). Rapidly transfer 0.090-0.110 g (weighed to the nearest 0.0001 g) of reagent-grade $Na_2C_4H_4O_6$ $2H_2O$ to the titration vessel.

b. Titrate to an endpoint in the same manner as with the sample (see 4.7.3.3).

c. Repeat the standardization procedure until three successive results agree within five parts per thousand.

d. If the indicated water equivalent (A) of the Karl Fischer reagent is less than 0.0014 g of water per ml of Karl Fischer reagent, it may be due to the presence of too much water in the absolute methanol used. In this case, distill the methanol from metallic calcium or calcium hydride. Passing the methanol through a column of Molecular Sieves, Type 4A, may also reduce the water content of the methanol sufficiently.

$$\text{Water equivalent (A)} = \frac{0.1566W}{V}$$

Where: A = water equivalent of the Karl Fischer reagent, g/ml

W = weight of $Na_2C_4H_4O_6$ $2H_2O$

V = volume of Karl Fischer reagent used, ml

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4.7.3.2.2 Water-in-methanol solution. The water-in-methanol solution should contain 0.0015-0.0020 g of water per ml of solution. A good grade of commercial absolute methanol contains about 0.0010 g water per ml of methanol. Water content can be adjusted by adding 1.0 g of water to 1000 ml of the water-in-methanol solution to produce a change of 0.0010 g per ml. Determine the relative strength of the water-in-methanol solution in terms of Karl Fischer reagent as follows:

a. Put about 50 ml of the anhydrous methanol used in 4.7.3.2a into the titration beaker of the Aquameter. Add a slight excess of Karl Fischer reagent (4.7.3.2a), then back-titrate with water-in-methanol solution (4.7.3.2b).

b. Then run in additional 5 to 8 ml of Karl Fischer reagent, read to the nearest 0.01 ml, and again back-titrate with water-in-methanol solution (read to the nearest 0.01 ml).

c. Repeat the addition and back-titrating steps twice more to provide triplicate determinations of the equivalency ratio.

d. Calculate the ratio (B) of Karl Fischer reagent to that of the water-in-methanol solution. The range of the ratios calculated from the three titrations should be not greater than 0.04. If the range exceeds 0.04, continue making titrations until three ratios are obtained whose range does not exceed 0.04.

e. Then determine the average ratio from all the ratios which have been obtained.

4.7.3.3 Procedure. Determine the moisture content of the sample by the Karl Fischer method using a direct-titration technique. Introduce to the titration beaker, through the opening in the diaphragm, approximately 10 g of sample weighed to the nearest 0.001 g and a mixture of 150 ml of anhydrous methanol with 50 ml of glacial acetic acid. Both materials are to be reagent grade. Close the opening, start the stirrer, and press the titrate button to titrate with Karl Fischer reagent (4.7.3.2a). When the indicator light glows,

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read the Karl Fischer burette to the nearest 0.01 ml. Where necessary, water-in-methanol solution (4.7.3.2b) may be used to back-titrate.

$$\text{Percent moisture} = \frac{100A\ (V_{KF} - BV_{WM})}{W}$$

Where: A = weight of water equivalent to 1.00 ml of Karl Fischer reagent, g/ml

V_{KF} = volume of Karl Fischer reagent titrant used, ml

V_{WM} = volume of water-in-methanol solution titrant used, ml

B = ratio of Karl Fischer reagent to that of water-in-methanol solution, ml/ml

W = weight of sample taken, g

4.7.4 Acceptance criteria. For the lot represented to pass the moisture test, the value obtained for the percent moisture shall be not greater than the value specified in 3.2.

4.8 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-1190 and MIL-STD-129.

5. PACKAGING

5.1 Preservation, Packaging, and Packing. Unless otherwise specified by the contract or purchase order, preservation, packaging, and packing requirements for the specified levels of protection shall be in accordance with MIL-STD-1190.

5.2 Marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked and labeled in accordance with the requirements of MIL-STD-129. If the material has toxic characteristics, the proper toxicity precautions shall be listed on the package.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Tetraethylenepentamine is intended for use as a catalyst in an epoxy potting compound for humidity and pressure seals.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Whether a first article is required (see 3.1)

d. Type and size of shipping container (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definition of "batch". A "batch" is defined as that quantity of material which has been subjected to one or more chemical or physical processes (or combinations thereof) intended to produce a desired product having substantially uniform characteristics.

6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.6 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Subject term (keyword) listing.

Catalysts
Chemical compounds
Encapsulants
Potting compounds

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Civilian Coordinating Activity
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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-47064A (MI)	2. DOCUMENT DATE (YYMMDD) 910930

3. DOCUMENT TITLE
TETRAETHYLENEPENTAMINE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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